

Simon Jones

Rory Solomon

Code as a Liberal Art

May 12, 2025

### YouTube Recommends

Last month was the 20th anniversary of YouTube’s first video. Uploaded on April 23, 2005 by one of YouTube’s founding engineers, Jawed Karim, the 18-second clip is titled “Me at the zoo,”<sup>[21]</sup> and features the man himself at the San Diego zoo, speaking to the camera whilst standing in front of the elephants. YouTube’s original trio of developers—Karim, Steven Chen, and Chad Hurley—built the site so users could easily share videos with one another, and view them from anywhere with access to the internet. But what started as a simple means of achieving easier connection would soon be at the forefront of a massive shift in internet culture and social behavior.

In the first year running, YouTube.com amassed over 25 million videos, with more than 20,000 videos being uploaded daily.<sup>[18]</sup> Within two months of its official launch at the end of 2005, roughly 15 million videos were being watched on YouTube per day.<sup>[44]</sup> Google bought the platform for roughly \$1.65 billion in late 2006, and as of 2025, over 2.6 million videos are uploaded to YouTube every single day—500 hours of content uploaded every minute—and the site averages 20 million daily users. (Adavelli, Stahl, Iqbal)

For the first years of YouTube, they didn’t use any kind of recommendation algorithm to push videos. Instead, the videos on YouTube’s homepage were manually chosen by a group of editors entitled “community managers,” mostly referred to as the “coolhunters.” (Bergen) This

team was charged with the task of perusing YouTube's corpus of content to find a diverse collection of videos to post on YouTube's homepage, updated throughout the day. Almost unfathomable to internet users today, for three years, the YouTube homepage looked the same for every person accessing the site. In 2007, both a "most viewed" and a "most liked" tab were added to the homepage, but it wasn't until 2009 that a "recommended videos" box was introduced, and another two years for that recommendation box to become the majority of the YouTube homepage.

YouTube is considered one of the first online streaming platforms, leading a major shift in how individuals viewed content at home. It was only two years after YouTube went live that Netflix shifted from mail-ordered DVD rentals to online streaming, (available for free for those already subscribed to the rental service.) Hulu followed shortly after, officially launching in 2008 along with Amazon Video On Demand (now Amazon Prime Video). Amazon originally had a service called Amazon Unbox which provided access to a large library of movies and television shows available for download over the internet, but Amazon Video On Demand provided content through online streaming rather than downloading. In 2010, Netflix started running their streaming service as its own platform and began producing original content in 2013 with their debut, *House of Cards*. From then, a gradual shift from cable TV to online streaming would take place until the streaming boom in 2019 and beyond, commonly referred to as the "streaming wars." The battle of streaming platforms has been progressing over the course of this century, but 2019 marked a turning point in the rise of streaming platforms. Apple TV+ and Disney+ both launched that year, HBOMax and Peacock launched in 2020, and Paramount+ and Discovery+ launched in 2021, along with a substantial amount of other streaming services, as more people made the decision to "cut the cord," aided largely by the 2020 pandemic. July 2022 marked the

first month where streaming surpassed cable television in viewers. (Nielsen) But what exactly is streaming in the first place?

Before streaming, television was strictly broadcasted over the airwaves. You could purchase a physical copy of a show or movie, and if you had access to an internet connection, there were services to download movies and TV shows, but this usually would take hours, even sometimes up to a day depending on the file size and your connection's bandwidth. Otherwise, you were at the will of the broadcast, where the shows you'd watch were transmitted to your home through radio signals: video was transmitted over AM and audio was FM. (Silva)

Streaming, however, is done entirely over the internet. Instead of downloading an entire file onto your device and playing it locally, streaming lets video play in a browser. Like everything on the internet, audio and video data is broken down into smaller chunks, called packets, which are encoded and compressed before the data leaves the server, and then decoded once it reaches the user's device. (Rogers) YouTube also uses HTTP-based adaptive bitrate streaming which transcodes video data to adapt it for different device limitations and internet strength so that users can get the highest quality content possible in the shortest time. Buffering is also a tool used to help load videos across different bandwidths and internet speeds, where the media player loads the video a few seconds ahead.

The first movie ever was streamed over the internet in 1993. It was the cult-followed *Wax: Or the Discovery of Television Among the Bees*. According to a Times article, it took up until almost halfway through the digital premiere for engineers to actually find and then transmit the internet signal playing the movie onto their computers. The color-image was reduced to black and white, displayed in an 8-inch window, the audio cutting in and out whenever the internet was overloaded with data traffic, and playing at a halting 2 frames per second, instead of the 24 frame

rate of a typical broadcast. (Markoff) While it would take the rest of the decade to improve streaming's quality, the new technology proved extremely valuable, and thus live-streaming became the first mainstream use of streaming technology. The first livestream happened about a month after the fiasco of *Wax*; a concert of the American rock band Severe Tire Damage. The drummer of the band, Mark Weiser, worked at the Xerox Palo Alto Research Center, where the first streaming technology was being developed. Testing out new streaming software Mbone, the band decided to stream their concert, which would then lead to The Rolling Stones wanting to livestream their concerts the following year. These livestreams were only able to reach a few hundred computers at the time, but still proved to be extremely worthwhile. In 1995, RealNetworks held the first ever public livestream, a baseball game between the New York Yankees and the Seattle Mariners. Then, in 1999, the first presidential webcast was streamed from George Washington University.

Even before YouTube came along, MLB.TV started streaming baseball games over the internet as early as 2002. The debut stream was of the August 26, 2002 game between the New York Yankees and the Texas Rangers and amassed 30,000 viewers. Livestreaming would not become part of YouTube's platform until 2008, but the technology behind it is what ended up being used to engineer the video-sharing platform.

Like many others of my generation, I've been using YouTube for most of my life. A website I'd go to for music and lyric videos as a kid quickly grew into this mega-social media platform in the early 2010s, with the help of monetization. In 2007, YouTube introduced the YouTube Partner Program (YPP), a new feature where channels meeting a certain criteria could take a percentage of the revenue coming from the ads playing on their videos. Thus began the long-winded spiral into partial damnation. I exaggerate, of course, but a shift definitely started

happening once creators could start making money from the content they uploaded, which in turn required more scrutiny on YouTube's behalf to make sure videos met advertising standards. For a few years, YouTube was a creative, indie online community, where individuals would create and share content with viewers in a more personal and freeform style than anyone was ever used to before. Moderation and scrutinization of content was loose and unrestrictive. It was so new at the time, so unprecedented, and best of all: it was completely free.

But regulation of material began happening even before the YPP. YouTube managed to avoid any copyright issues for its first few months, despite a lot of copyrighted material being uploaded to the platform, but that all changed by 2006, after the first television clip went viral on the platform. On December 17, 2005, Saturday Night Live premiered their first ever digital short, entitled "Lazy Sunday." It aired at the very end of the episode, and the only way to watch it after its airing was on YouTube. (If not for YouTube there might have been no way to watch the clip after the fact.) Within just a few days, the bootleg video of the skit amassed over 2 million views, and was responsible for bringing a ton of new users to the platform and boosting YouTube into the public eye. On February 16, 2006, YouTube made a post to their community explaining that NBC has asked them to remove the video from the platform; "We know how popular that video is but YouTube respects the rights of copyright holders." (The Youtube Team) Funnily enough, today you can still watch the skit on YouTube, uploaded on SNL's own YouTube channel.

Copyright infringement would become a huge aspect of YouTube's content guidelines, becoming increasingly more restrictive overtime. I remember when I was first watching YouTube that playing music of any sort in your video was not a serious issue; today, even humming a few seconds of a song could result in a copyright strike on the platform.

So finding videos on YouTube used to come down to manual searches, most-viewed content, and whatever the coolhunters scrounged up that day. But as the mass of content on YouTube became larger than one could imagine, (today, there are roughly 5.1 billion videos on the platform, (SEO)) it became clear that they would need a more efficient way to help users find videos they were most likely to enjoy. So the coolhunters were phased out and relocated in favor of implementing a recommendation algorithm, which is now how the entirety of the YouTube homepage is produced for each individual user. This recommendation algorithm uses a combination of video metadata and user history to curate a selection of videos they think each individual is most likely to click on. Metadata includes things like title, video description, length, view count, and where it was uploaded from. YouTube uses this contextual data to sort the video into the most accurate category. Then, analyzing what videos in one's watch history are from which algorithmically assigned topics, the recommendation system can suggest other videos from the topics they think the user is most interested in. But another important aspect of the YouTube algorithm is association rule mining, or co-visitation counts. This means that the algorithm actually compares one's watch history to the watch history of other users. The algorithm will find videos to recommend based on other users with similar interests, a model of if you like this, then you'll like this. So the algorithm will recommend channels or videos to users if those channels and videos tend to have an overlap in viewers; based on who does and doesn't like a recommended video, YouTube will then know who to recommend it to by comparing other user's watch histories to the histories of the users who have liked the new video. Even if two videos have no topic alignment, YouTube can confidently recommend such videos to other users. This helps with one of YouTube's main goals in recommending videos: a diversity of topics and creators to reflect the vast range of material on the platform. They do this by limiting the amount

of recommendations generated by a single topic or channel in one's watch history. At the same time that the algorithm is analyzing your watch history in comparison to others and recommending videos based on that, it is also figuring out *what the video even is* based on how users are interacting with it. The algorithm uses machine learning both to improve its own ability to determine more detailed recommendations and also to get better at accurately sorting the massive corpus of videos.

YouTube used to optimize content engagement based on whether or not a user clicks on a video, because every click counted as a view for the video: clicking was a success for the algorithm. But this allowed users to figure out how to exploit the algorithm, mostly with the overabundance of clickbait, which led to the algorithm being more meticulous and using watch time as a major factor in generating recommendations. High click rates but low watch times would signify to the algorithm that the metadata of the video may be misleading, or that the video's quality may be sub par. But this ultimately led to an imbalance when it came to video length, as naturally a longer video would have a longer watch time than a shorter video, which didn't necessarily reflect the quality of the video. This resulted in an influx of long form content being created, such as gaming videos. So the algorithm got more complicated to accommodate such changes, both by developers and on its own, and by 2016, it was seeing a massive shift in going from a recommender to a predictor. As it was originally designed, the algorithm was only generating recommendations based on user and meta data, but as it focused more on successful clickthrough rates, it began to function by predicting what users might do, a.k.a if a user would click and watch a video all the way through. This became the algorithms main focus more than just trying to provide a good range of options to satisfy a users watch desires at any given

moment. This shift was also largely influenced by advertising on the platform and more meticulous guidelines around age appropriate content.

While the YouTube algorithm has not been overly accused of implicit bias, as recommendations are largely based on what individuals are watching and not the content or popularity of videos alone, the platform has come under scrutiny many times for biases around content restriction, which are implemented both algorithmically and by human reviewers at the company. There was a large scandal in 2017 where YouTube had to publicly apologize for an update in its age-restricted content analyzer for misappropriating an abundance of LGBTQ+ content as not suitable for children. This doesn't just block a lot of content from users, but can make it more difficult to search for it, removes from any trending pages, and can place specific users under more harsh scrutiny with future uploads. Some creators tested the issue, uploading the same exact video with different titles and seeing only the ones with queer language in them (like "trans" and "gay") got age-restricted while the others didn't. While YouTube promised to correct this, in 2019, a suit was filed against the company accusing it of discrimination. YouTube won the case, and the issue has seemed to be largely resolved as of 2025, but that hasn't stopped users from marginalized groups from voicing complaints that they feel their content is more algorithmically challenged on the platform.

Furthermore, YouTube has come under harsh scrutiny for its promotion of problematic content, like racist videos or harmful conspiracy videos. And while YouTube has voiced a commitment to disallow "borderline content," or content that doesn't breach but toes the line of their community agreement, from being pushed by the recommendation system, it hasn't stopped a lot of alt-right content from cropping up in users recommendations. One research study analyzing the likelihood of being recommended such content found that "a user who pursues even



mildly conservative content is ‘only one or two clicks away from extreme far-right channels, conspiracy theories, and radicalizing content.’” (Bryant) While this is really concerning, it does make sense given the nature of the algorithm. If that is the type of content users viewing one type of video watch, then if you also start watching that one type of video, you’ll start getting recommended the other stuff too. It’s more a question of how YouTube can filter content better within its recommendation algorithm than a question of the algorithm's bias itself. But findings have shown that YouTube is more likely to recommend controversial and conservative content, and studies have suggested that YouTube be more transparent about their algorithm for further studying, to attempt to derive solutions where such trends could be avoided going forward.

This concept makes up the entire nature of Ruha Benjamin’s book, “Race After Technology,” which looks at how code and algorithms perpetuate racial stereotypes. While YouTube has managed to avoid such scrutinization, it’s good to question things like why you only get recommended content from creators of a specific race, why YouTube’s “diversity” of topics and channels doesn’t really equate necessarily to a true diverse array of perspectives, instead more likely to recommend different channels all promoting the same perspective, because that is what you were watching already. It’s rare that I find a new form of content entirely on YouTube. From personal experience of interacting with this recommendation very intimately, for over 10 years, and with YouTube being overwhelmingly my most used website and application all that time, the recommendations feel more like my comfort zone than any sort of attempt to broaden what I watch. That being said, I have also stumbled upon conservative content, which I never engage with. If I leave autoplay on and fall asleep, I will wake with a video playing that I *never* would’ve intentionally clicked, and sometimes the right-leaning podcasts. Even directly telling the platform that I don’t like such content hasn’t stopped similar stuff from coming up again. It’s difficult to

understand the true intentions and functionality of a platform's algorithm when they don't actually provide all the information about how it works, instead making us infer from our own experience with and from research papers over a decade old which no longer encompass all the newer aspects of the system.

The Bryant research study warns of the dangers of allowing the recommendation system to continue function as it does, where users absorbing only one kind of content and thus being only recommend content in that vein can isolate themselves from other views and ideas and nurture a skewed idea of the world and the complexity and diversity of people's views and understandings. "There is danger in allowing our media to define a political spectrum that is not representative of the actual country." In allowing "a vocal minority to be represented as a majority," and thus skew the actuality of the public's consensus. (Bryant) But the best way to combat this is to always be skeptical of recommendation systems, always wonder why certain content may be pushed more than others, and what that other content might actually be. I will go out of my way to try and tailor my algorithms so that they actually reflect the diverse range of interests I have. Ultimately, while we all like our recommendation systems, especially when they are accurate, they shouldn't be the exclusive method we use to determine the media that influences us.

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